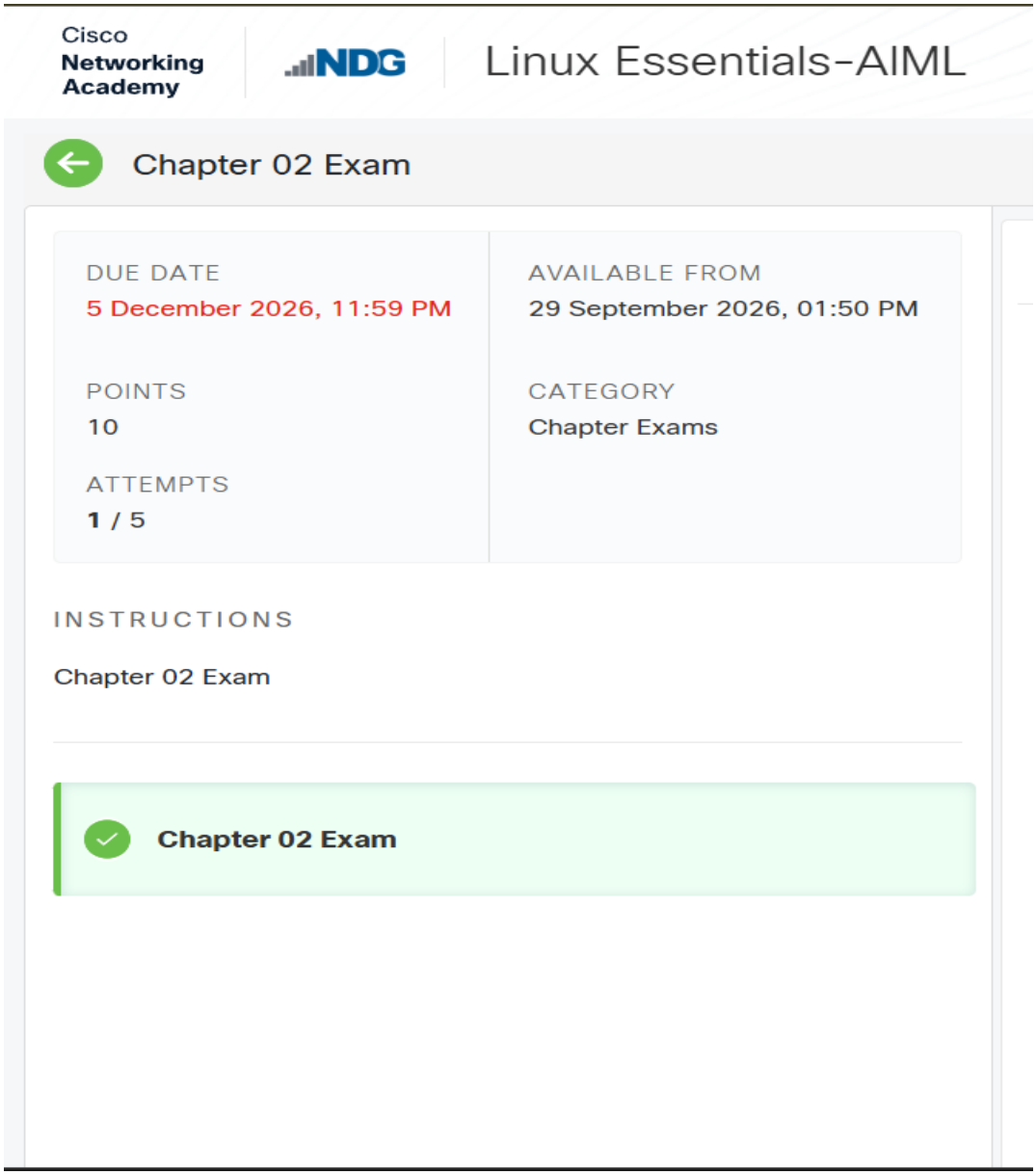
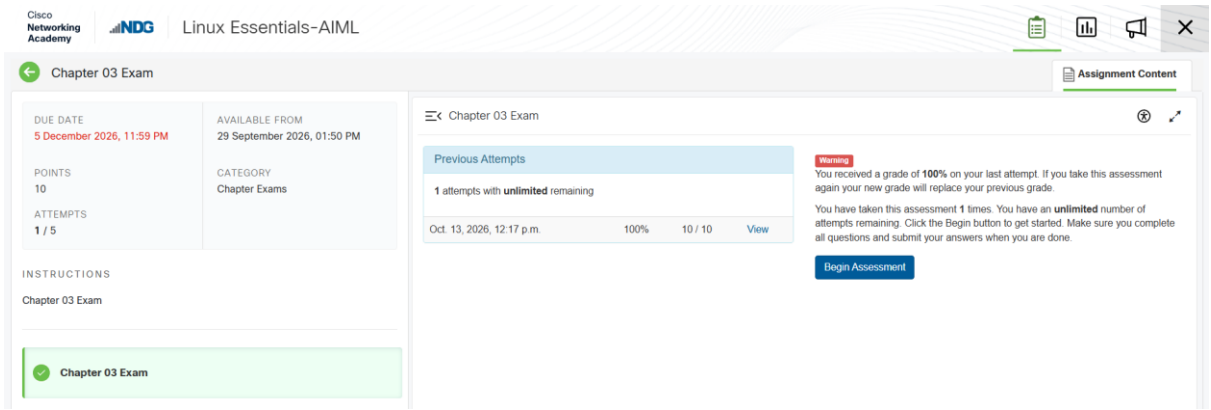
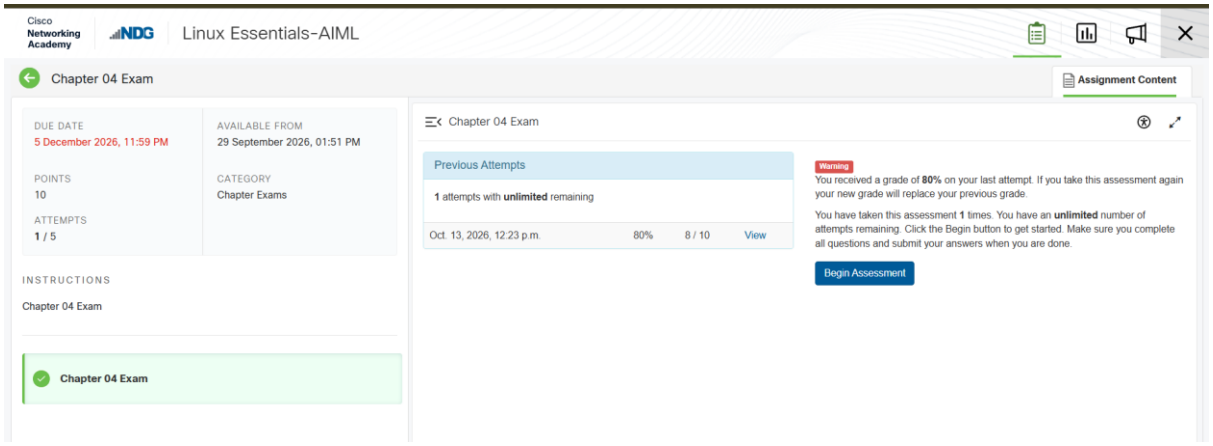
Chapter 2 :



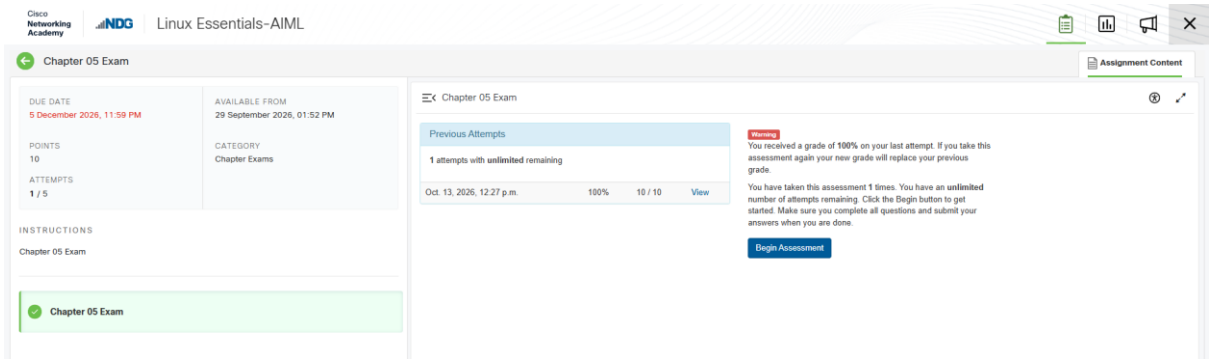
Chapter 3 :



Chapter 4 :



Chapter 5 :



LAB-5 :

```
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 6.12.90+deb13-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.


This lab has two user accounts (username :: password )

   root     :: netlab123
   sysadmin :: netlab123

Press the [Enter] key to begin...

sysadmin@localhost:~$
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

```
total 4
drwxr-xr-x 2 sysadmin sysadmin    6 Feb  8  2021 Desktop
drwxr-xr-x 4 sysadmin sysadmin 4096 Feb  8  2021 Documents
drwxr-xr-x 2 sysadmin sysadmin    6 Feb  8  2021 Downloads
drwxr-xr-x 2 sysadmin sysadmin    6 Feb  8  2021 Music
drwxr-xr-x 2 sysadmin sysadmin    6 Feb  8  2021 Pictures
drwxr-xr-x 2 sysadmin sysadmin    6 Feb  8  2021 Public
drwxr-xr-x 2 sysadmin sysadmin    6 Feb  8  2021 Templates
drwxr-xr-x 2 sysadmin sysadmin    6 Feb  8  2021 Videos
sysadmin@localhost:~$ ls -l /home
total 0
drwxr-xr-x 1 sysadmin sysadmin 28 Oct 12 13:50 sysadmin
sysadmin@localhost:~$ whoami
sysadmin
sysadmin@localhost:~$
sysadmin@localhost:~$ uname
Linux
sysadmin@localhost:~$ uname -n
localhost
sysadmin@localhost:~$ uname --nodename
localhost
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$ history
```

```
localhost
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$ history
    1  ls
    2  ls -l
    3  ls -l /home
    4  whoami
    5  uname
    6  uname -n
    7  uname --nodename
    8  pwd
    9  history
sysadmin@localhost:~$ history 5
    6  uname -n
    7  uname --nodename
    8  pwd
    9  history
   10  history 5
sysadmin@localhost:~$ !9
history
    1  ls
    2  ls -l
    3  ls -l /home
```

```
    8  pwd
    9  history
   10  history 5
sysadmin@localhost:~$ !9
history
    1  ls
    2  ls -l
    3  ls -l /home
    4  whoami
    5  uname
    6  uname -n
    7  uname --nodename
    8  pwd
    9  history
   10  history 5
   11  history
sysadmin@localhost:~$ echo Hello Student
Hello Student
sysadmin@localhost:~$
sysadmin@localhost:~$ echo $HISTSIZE
1000
sysadmin@localhost:~$ echo $PATH
/home/sysadmin/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:
```

```
sysadmin@localhost:~$
sysadmin@localhost:~$ Which date

Command 'Which' not found, did you mean:

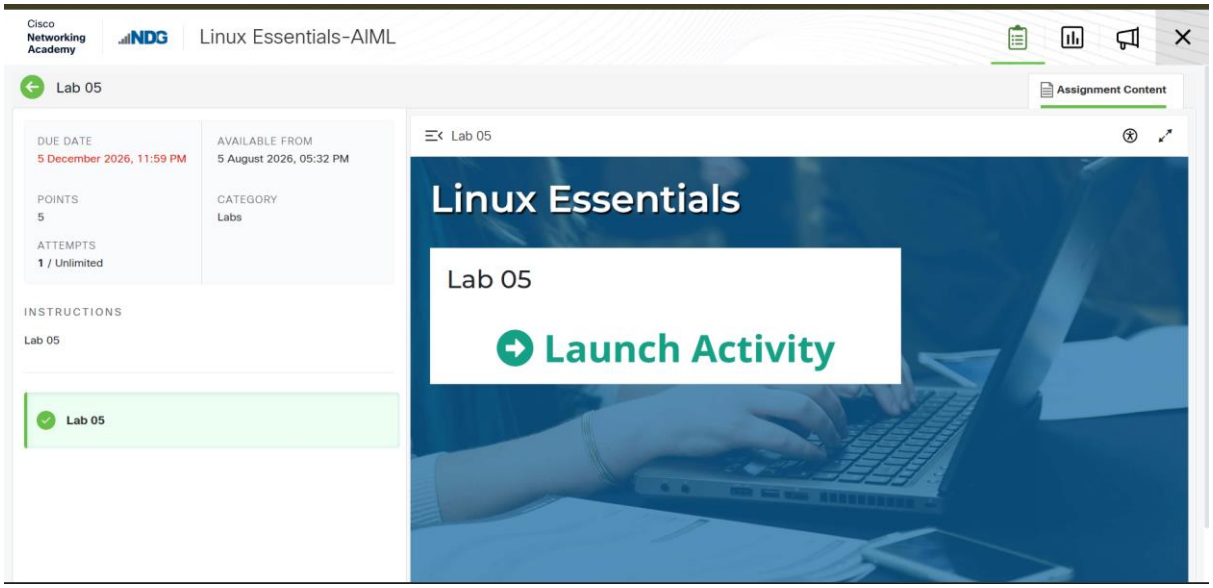
  command 'which' from deb debianutils

Try: sudo apt install <deb name>

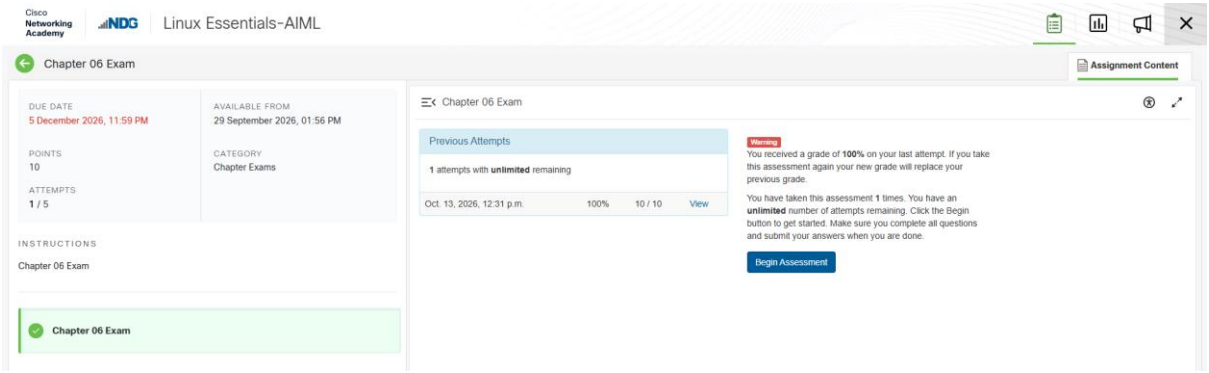
sysadmin@localhost:~$ which date
/bin/date
sysadmin@localhost:~$ type cd
cd is a shell builtin
sysadmin@localhost:~$ which ls
/bin/ls
sysadmin@localhost:~$ type cp
cp is /bin/cp
sysadmin@localhost:~$ which cp
/bin/cp
sysadmin@localhost:~$ type -a ls
ls is aliased to `ls --color=auto'
ls is /bin/ls
sysadmin@localhost:~$ alias
alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo terminal || echo
```

```
ls is aliased to `ls --color=auto'
ls is /bin/ls
sysadmin@localhost:~$ alias
alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo terminal || echo
 error)" "$(history|tail -n1|sed -e '\''s/^\s*[0-9]\+\s*//;s/[;&|]\s*alert$//'\'
')"'
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'
sysadmin@localhost:~$ type vi
vi is /usr/bin/vi
sysadmin@localhost:~$ cd /bin
sysadmin@localhost:/bin$ type vlc
-bash: type: vlc: not found
sysadmin@localhost:/bin$ cd
sysadmin@localhost:~$
sysadmin@localhost:~$ echo tody is 'date'
tody is date
sysadmin@localhost:~$ echo today is $(date)
today is Mon Oct 12 14:11:06 UTC 2026
```

```
sysadmin@localhost:~$ echo this is the command \`date\`
this is the command `date`
sysadmin@localhost:~$ echo this is the command "`date`"
this is the command Mon Oct 12 14:13:37 UTC 2026
sysadmin@localhost:~$ echo D*
Desktop Documents Downloads
sysadmin@localhost:~$ echo "D*"
D*
sysadmin@localhost:~$ echo Hello; echo Linux; echo Student
Hello
Linux
Student
sysadmin@localhost:~$ false; echo Not; echo Conditional
Not
Conditional
sysadmin@localhost:~$ echo start && echo Going && echo gone
start
Going
gone
sysadmin@localhost:~$ echo Success && false && echo Bye
Success
sysadmin@localhost:~$ false || echo fail or
fail or
sysadmin@localhost:~$
```



Chapter 6 :



LAB-6 :

```
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 6.12.90+deb13-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.


This lab has two user accounts (username :: password )

   root     :: netlab123
   sysadmin :: netlab123

Press the [Enter] key to begin...

sysadmin@localhost:~$ date
Mon Oct 12 14:38:59 UTC 2026
sysadmin@localhost:~$ man date
```

```
sysadmin@localhost:~$ date
Mon Oct 12 14:38:59 UTC 2026
sysadmin@localhost:~$ man date
sysadmin@localhost:~$ man -k password
chage (1)            - change user password expiry information
chgpasswd (8)        - update group passwords in batch mode
chpasswd (8)         - update passwords in batch mode
cpgr (8)             - copy with locking the given file to the password or gr...
cppw (8)             - copy with locking the given file to the password or gr...
crypt (3)            - password and data encryption
crypt_r (3)          - password and data encryption
endpwent (3)         - get password file entry
endspent (3)         - get shadow password file entry
expiry (1)           - check and enforce password expiration policy
fgetpwent (3)        - get password file entry
fgetspent (3)        - get shadow password file entry
fgetspent_r (3)      - get shadow password file entry
getpass (3)          - get a password
getpw (3)            - reconstruct password line entry
getpwent (3)         - get password file entry
getpwnam (3)         - get password file entry
getpwnam_r (3)       - get password file entry
getpwuid (3)         - get password file entry
```

```
grpunconv (8)        - convert to and from shadow passwords and groups
lckpwdf (3)          - get shadow password file entry
login.defs (5)       - shadow password suite configuration
openssl-passwd (1ssl) - compute password hashes
openssl-srp (1ssl)   - maintain SRP password file
pam_pwhistory (8)    - PAM module to remember last passwords
pam_unix (8)         - Module for traditional password authentication
passwd (1)           - change user password
passwd (1ssl)        - compute password hashes
passwd (5)           - the password file
passwd2des (3)       - RFS password encryption
putpwent (3)         - write a password file entry
putspent (3)         - get shadow password file entry
pwck (8)             - verify integrity of password files
pwconv (8)           - convert to and from shadow passwords and groups
pwunconv (8)         - convert to and from shadow passwords and groups
setpwent (3)         - get password file entry
setspent (3)         - get shadow password file entry
sgetspent (3)        - get shadow password file entry
sgetspent_r (3)      - get shadow password file entry
shadow (5)           - shadowed password file
shadowconfig (8)     - toggle shadow passwords on and off
srp (1ssl)           - maintain SRP password file
systemd-ask-password (1) - Query the user for a system password
```

```
shadow (5)              - shadowed password file
shadowconfig (8)        - toggle shadow passwords on and off
srp (1ssl)              - maintain SRP password file
systemd-ask-password (1) - Query the user for a system password
systemd-ask-password-console.path (8) - Query the user for system passwords o...
systemd-ask-password-console.service (8) - Query the user for system password...
systemd-ask-password-wall.path (8) - Query the user for system passwords on t...
systemd-ask-password-wall.service (8) - Query the user for system passwords o...
systemd-tty-ask-password-agent (1) - List or process pending systemd password...
ulckpwdf (3)            - get shadow password file entry
unix_chkpwd (8)         - Helper binary that verifies the password of the curren...
unix_update (8)         - Helper binary that updates the password of a given user
vigr (8)                - edit the password, group, shadow-password or shadow-gr...
vipw (8)                - edit the password, group, shadow-password or shadow-gr...
xcrypt (3)              - RFS password encryption
xdecrypt (3)            - RFS password encryption
xencrypt (3)            - RFS password encryption
sysadmin@localhost:~$
sysadmin@localhost:~$ apropos password
chage (1)               - change user password expiry information
chgpasswd (8)           - update group passwords in batch mode
chpasswd (8)            - update passwords in batch mode
cpgr (8)                - copy with locking the given file to the password or gr...
cppw (8)                - copy with locking the given file to the password or gr...
```

```
chgpasswd (8)           - update group passwords in batch mode
chpasswd (8)            - update passwords in batch mode
cpgr (8)                - copy with locking the given file to the password or gr...
cppw (8)                - copy with locking the given file to the password or gr...
crypt (3)               - password and data encryption
crypt_r (3)             - password and data encryption
endpwent (3)            - get password file entry
endspent (3)            - get shadow password file entry
expiry (1)              - check and enforce password expiration policy
fgetpwent (3)           - get password file entry
fgetspent (3)           - get shadow password file entry
fgetspent_r (3)         - get shadow password file entry
getpass (3)             - get a password
getpw (3)               - reconstruct password line entry
getpwent (3)            - get password file entry
getpwnam (3)            - get password file entry
getpwnam_r (3)          - get password file entry
getpwuid (3)            - get password file entry
getpwuid_r (3)          - get password file entry
getspent (3)            - get shadow password file entry
getspent_r (3)          - get shadow password file entry
getspnam (3)            - get shadow password file entry
getspnam_r (3)          - get shadow password file entry
grpconv (8)             - convert to and from shadow passwords and groups
```

```
setpwent (3)          - get password file entry
setspent (3)          - get shadow password file entry
sgetspent (3)         - get shadow password file entry
sgetspent_r (3)       - get shadow password file entry
shadow (5)            - shadowed password file
shadowconfig (8)      - toggle shadow passwords on and off
srp (1ssl)            - maintain SRP password file
systemd-ask-password (1) - Query the user for a system password
systemd-ask-password-console.path (8) - Query the user for system passwords o...
systemd-ask-password-console.service (8) - Query the user for system password...
systemd-ask-password-wall.path (8) - Query the user for system passwords on t...
systemd-ask-password-wall.service (8) - Query the user for system passwords o...
systemd-tty-ask-password-agent (1) - List or process pending systemd password...
ulckpwdf (3)          - get shadow password file entry
unix_chkpwd (8)       - Helper binary that verifies the password of the curren...
unix_update (8)       - Helper binary that updates the password of a given user
vigr (8)              - edit the password, group, shadow-password or shadow-gr...
vipw (8)              - edit the password, group, shadow-password or shadow-gr...
xcrypt (3)            - RFS password encryption
xdecrypt (3)          - RFS password encryption
xencrypt (3)          - RFS password encryption
sysadmin@localhost:~$ man -f passwd
passwd (1ssl)         - compute password hashes
passwd (1)            - change user password
```

```
xencrypt (3)          - RFS password encryption
sysadmin@localhost:~$ man -f passwd
passwd (1ssl)         - compute password hashes
passwd (1)            - change user password
passwd (5)            - the password file
sysadmin@localhost:~$ man 5 passwd
sysadmin@localhost:~$ whatis passwd
passwd (1ssl)         - compute password hashes
passwd (1)            - change user password
passwd (5)            - the password file
sysadmin@localhost:~$ info date
sysadmin@localhost:~$ date --help
Usage: date [OPTION]... [+FORMAT]
  or:  date [-u|--utc|--universal] [MMDDhhmm[[CC]YY][.ss]]
Display the current time in the given FORMAT, or set the system date.

Mandatory arguments to long options are mandatory for short options too.
  -d, --date=STRING          display time described by STRING, not 'now'
      --debug                annotate the parsed date,
                              and warn about questionable usage to stderr
  -f, --file=DATEFILE        like --date; once for each line of DATEFILE
  -I[FMT], --iso-8601[=FMT]  output date/time in ISO 8601 format.
                               FMT='date' for date only (the default),
                               'hours', 'minutes', 'seconds', or 'ns'
```

```
  -f, --file=DATEFILE        like --date; once for each line of DATEFILE
  -I[FMT], --iso-8601[=FMT]  output date/time in ISO 8601 format.
                               FMT='date' for date only (the default),
                               'hours', 'minutes', 'seconds', or 'ns'
                               for date and time to the indicated precision.
                               Example: 2006-08-14T02:34:56-06:00
  -R, --rfc-email            output date and time in RFC 5322 format.
                               Example: Mon, 14 Aug 2006 02:34:56 -0600
      --rfc-3339=FMT         output date/time in RFC 3339 format.
                               FMT='date', 'seconds', or 'ns'
                               for date and time to the indicated precision.
                               Example: 2006-08-14 02:34:56-06:00
  -r, --reference=FILE       display the last modification time of FILE
  -s, --set=STRING           set time described by STRING
  -u, --utc, --universal     print or set Coordinated Universal Time (UTC)
      --help     display this help and exit
      --version  output version information and exit

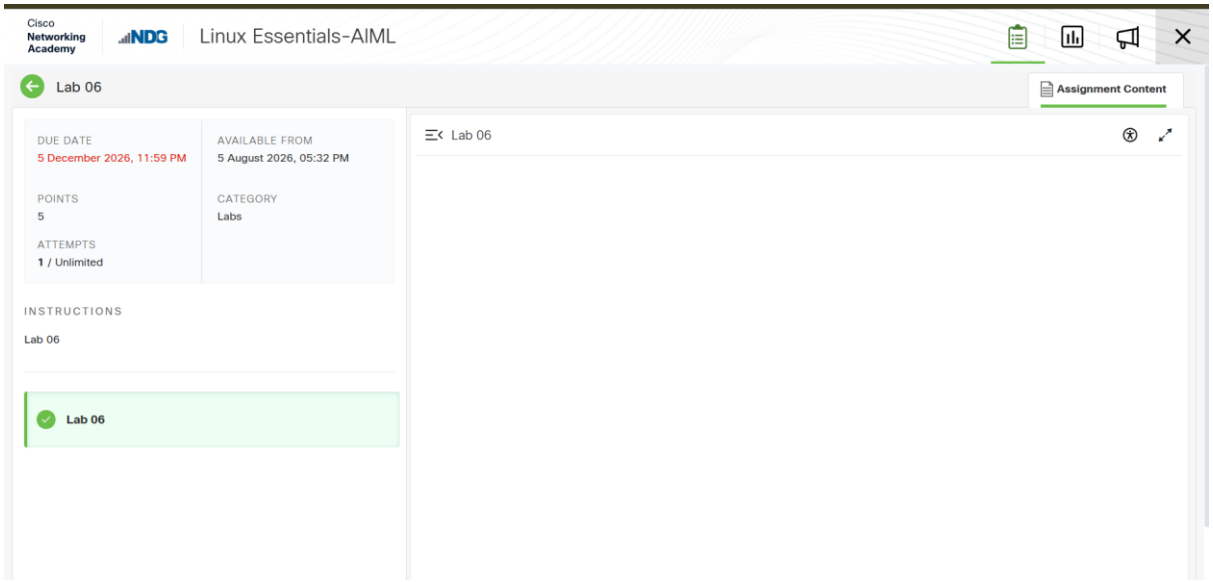
FORMAT controls the output.  Interpreted sequences are:

  %%   a literal %
  %a   locale's abbreviated weekday name (e.g., Sun)
  %A   locale's full weekday name (e.g., Sunday)
  %b   locale's abbreviated month name (e.g., Jan)
```

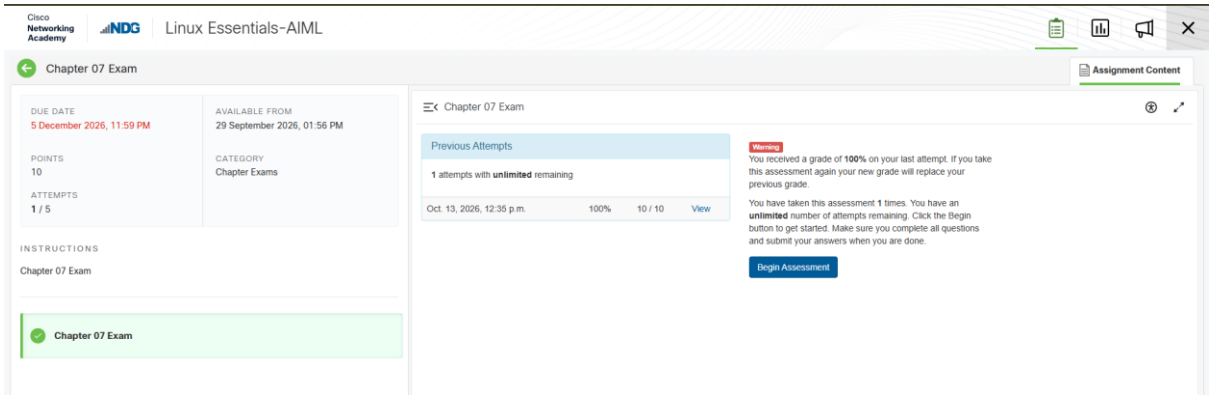
```
  %F   full date; same as %Y-%m-%d
  %g   last two digits of year of ISO week number (see %G)
  %G   year of ISO week number (see %V); normally useful only with %V
  %h   same as %b
  %H   hour (00..23)
  %I   hour (01..12)
  %j   day of year (001..366)
  %k   hour, space padded ( 0..23); same as %_H
  %l   hour, space padded ( 1..12); same as %_I
  %m   month (01..12)
  %M   minute (00..59)
  %n   a newline
  %N   nanoseconds (000000000..999999999)
  %p   locale's equivalent of either AM or PM; blank if not known
  %P   like %p, but lower case
  %q   quarter of year (1..4)
  %r   locale's 12-hour clock time (e.g., 11:11:04 PM)
  %R   24-hour hour and minute; same as %H:%M
  %s   seconds since 1970-01-01 00:00:00 UTC
  %S   second (00..60)
  %t   a tab
  %T   time; same as %H:%M:%S
  %u   day of week (1..7); 1 is Monday
  %U   week number of year, with Sunday as first day of week (00..53)
```

```
GNU coreutils online help: <http://www.gnu.org/software/coreutils/>
Report date translation bugs to <http://translationproject.org/team/>
Full documentation at: <http://www.gnu.org/software/coreutils/date>
or available locally via: info '(coreutils) date invocation'
sysadmin@localhost:~$ ls /usr/share/doc
accountsservice             libestr0                libxext6
adduser                     libevent-2.1-6          libxml2
apparmor                    libexpat1               libxmuu1
apt                         libext2fs               libxtables12
apt-utils                   libext2fs2              libyaml-0-2
ascii                       libfastjson4            libzstd1
base-files                  libfdisk1               linux-base
base-passwd                 libffi6                 linux-libc-dev
bash                        libfribidi0             locales
bash-completion             libfuse2                login
bind9                       libgcc-7-dev            logrotate
bind9-host                  libgcc1                 lsb-base
bind9utils                  libgcrypt20             lsb-release
binutils                    libgdbm-compat4         lshw
binutils-common             libgdbm5                lsof
binutils-x86-64-linux-gnu   libgeoip1              ltrace
bsdmainutils                libgirepository-1.0-1   man-db
bsdutils                    libglib2.0-0            manpages
busybox-initramfs           libglib2.0-data         manpages-dev
```

```
libcroco3                   libudev1                wamerican
libcryptsetup12             libunistring2           wget
libdb5.3                    libunwind8              whiptail
libdbus-1-3                 libusb-1.0-0            xauth
libdebconfclient0           libutempter0            xdg-user-dirs
libdevmapper1.02.1          libuuid1                xkb-data
libdns-export1100           libwrap0                xxd
libdns1100                  libx11-6                xz-utils
libdrm-common               libx11-data             zip
libdrm2                     libxau6                 zlib1g
libedit2                    libxcb1
libelf1                     libxdmcp6
sysadmin@localhost:~$ locate crontab
/etc/crontab
/usr/bin/crontab
/usr/share/bash-completion/completions/crontab
/usr/share/doc/cron/examples/crontab2english.pl
/usr/share/man/man1/crontab.1.gz
/usr/share/man/man5/crontab.5.gz
sysadmin@localhost:~$ locate -b /etc/crontab
sysadmin@localhost:~$ whereis passwd
passwd: /usr/bin/passwd /etc/passwd /usr/share/man/man1/passwd.1.gz /usr/share/m
an/man1/passwd.1ssl.gz /usr/share/man/man5/passwd.5.gz
sysadmin@localhost:~$
```


Cisco Networking Academy
NDG
Linux Essentials-AIML
Lab 06
Assignment Content
DUE DATE
5 December 2026, 11:59 PM
AVAILABLE FROM
5 August 2026, 05:32 PM
POINTS
5
CATEGORY
Labs
ATTEMPTS
1 / Unlimited
INSTRUCTIONS
Lab 06
Lab 06
Lab 06

Chapter 7 ：


Cisco Networking Academy
NDG
Linux Essentials-AIML
Chapter 07 Exam
Assignment Content
DUE DATE
5 December 2026, 11:59 PM
AVAILABLE FROM
29 September 2026, 01:56 PM
POINTS
10
CATEGORY
Chapter Exams
ATTEMPTS
1 / 5
INSTRUCTIONS
Chapter 07 Exam
Chapter 07 Exam
Chapter 07 Exam
Previous Attempts
1 attempts with unlimited remaining
Oct 13, 2026, 12:35 p.m. 100% 10 / 10 View
Warning
You received a grade of 100% on your last attempt. If you take this assessment again your new grade will replace your previous grade.
You have taken this assessment 1 times. You have an unlimited number of attempts remaining. Click the Begin button to get started. Make sure you complete all questions and submit your answers when you are done.
Begin Assessment

LAB-7 :

```
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 6.12.90+deb13-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.


This lab has two user accounts (username :: password )

   root     :: netlab123
   sysadmin :: netlab123

Press the [Enter] key to begin...

sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$ echo $HOME
```

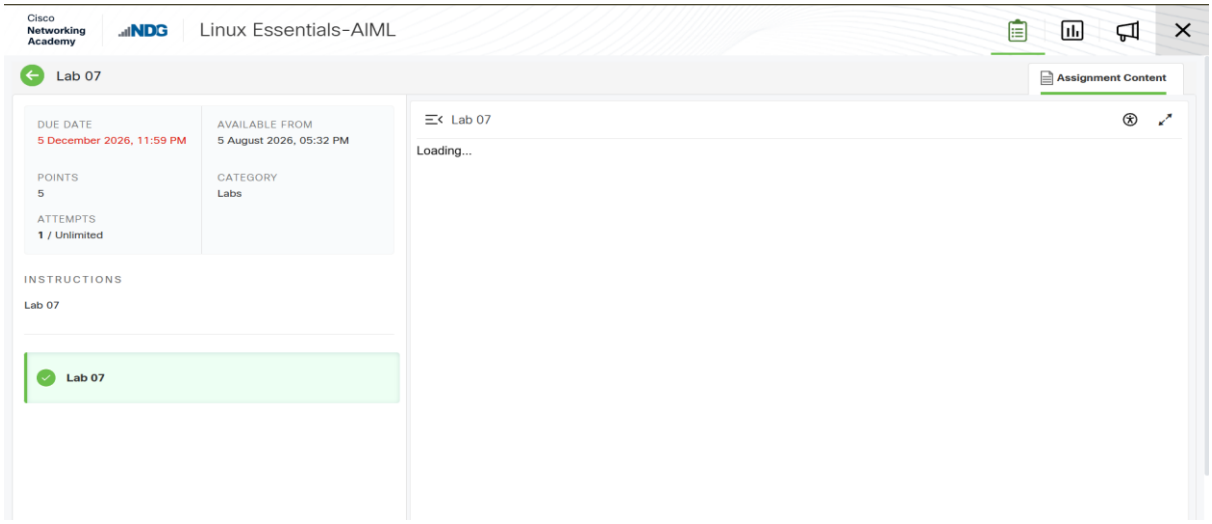
```
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$ echo $HOME
/home/sysadmin
sysadmin@localhost:~$ cd /
sysadmin@localhost:/$ pwd
/
sysadmin@localhost:/$ cd
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$ cd /home
sysadmin@localhost:/home$ pwd
/home
sysadmin@localhost:/home$ cd ~
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$ echo ~ ~sysadmin ~root ~mail ~nobody
/home/sysadmin /home/sysadmin /root /var/mail /nonexistent
sysadmin@localhost:~$ cd ~root
-bash: cd: /root: Permission denied
sysadmin@localhost:~$ cd /usr/bin
sysadmin@localhost:/usr/bin$ pwd
/usr/bin
```

```
sysadmin@localhost:/usr$ pwd
/usr
sysadmin@localhost:/usr$ cd /usr/share/doc
sysadmin@localhost:/usr/share/doc$ pwd
/usr/share/doc
sysadmin@localhost:/usr/share/doc$ cd bash
sysadmin@localhost:/usr/share/doc/bash$ pwd
/usr/share/doc/bash
sysadmin@localhost:/usr/share/doc/bash$ cd ..
sysadmin@localhost:/usr/share/doc$ pwd
/usr/share/doc
sysadmin@localhost:/usr/share/doc$ cd ../dict
sysadmin@localhost:/usr/share/dict$ pwd
/usr/share/dict
sysadmin@localhost:/usr/share/dict$ cd
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
sysadmin@localhost:~$ ls -a
.              .bashrc    .selected_editor           Documents  Pictures   Videos
..             .cache     .sudo_as_admin_successful  Downloads  Public
.bash_logout   .profile   Desktop                    Music      Templates
sysadmin@localhost:~$ ls -l /etc/hosts
-rw-r--r-- 1 root root 172 Oct 12 15:01 /etc/hosts
sysadmin@localhost:~$ ls -R /etc/udev
```

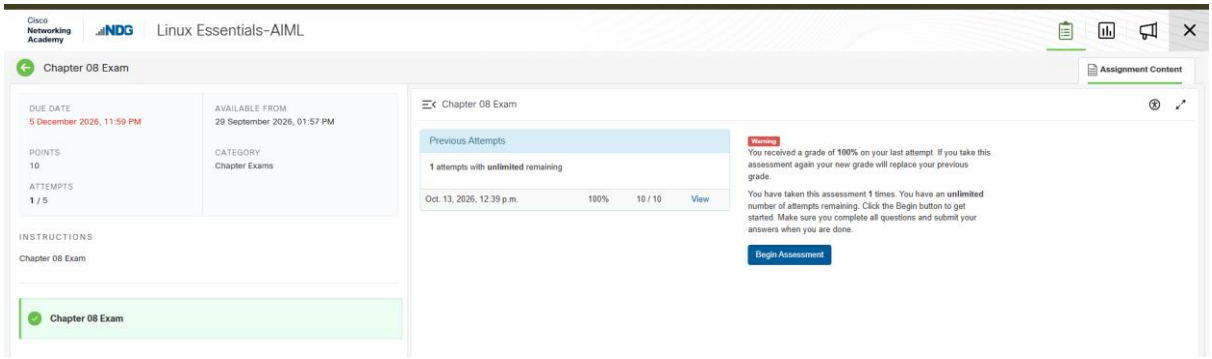
```
/etc/udev/hwdb.d:

/etc/udev/rules.d:
70-persistent-cd.rules  README
sysadmin@localhost:~$ ls -d /etc/s*
/etc/screenrc    /etc/services  /etc/skel    /etc/subgid-  /etc/sudoers.d
/etc/securetty   /etc/shadow    /etc/ssh     /etc/subuid   /etc/sysctl.conf
/etc/security    /etc/shadow-   /etc/ssl     /etc/subuid-  /etc/sysctl.d
/etc/selinux     /etc/shells    /etc/subgid  /etc/sudoers  /etc/systemd
sysadmin@localhost:~$ ls -d /etc/????
/etc/bind  /etc/dpkg  /etc/mtab  /etc/perl  /etc/udev
/etc/dhcp  /etc/motd  /etc/newt  /etc/skel
sysadmin@localhost:~$ ls -d /etc/[abcd]
ls: cannot access '/etc/[abcd]': No such file or directory
sysadmin@localhost:~$ ls -d /etd/[abcd]*
ls: cannot access '/etd/[abcd]*': No such file or directory
sysadmin@localhost:~$ ls -d /etc/[abcd]*
/etc/adduser.conf            /etc/binfmt.d              /etc/crontab
/etc/alternatives            /etc/ca-certificates       /etc/dbus-1
/etc/apm                     /etc/ca-certificates.conf  /etc/debconf.conf
/etc/apparmor                /etc/calendar              /etc/debian_version
/etc/apparmor.d              /etc/console-setup         /etc/default
/etc/apt                     /etc/cron.d                /etc/deluser.conf
```

```
/etc/securetty  /etc/shadow    /etc/ssh     /etc/subuid   /etc/sysctl.conf
/etc/security   /etc/shadow-   /etc/ssl     /etc/subuid-  /etc/sysctl.d
/etc/selinux    /etc/shells    /etc/subgid  /etc/sudoers  /etc/systemd
sysadmin@localhost:~$ ls -d /etc/????
/etc/bind  /etc/dpkg  /etc/mtab  /etc/perl  /etc/udev
/etc/dhcp  /etc/motd  /etc/newt  /etc/skel
sysadmin@localhost:~$ ls -d /etc/[abcd]
ls: cannot access '/etc/[abcd]': No such file or directory
sysadmin@localhost:~$ ls -d /etd/[abcd]*
ls: cannot access '/etd/[abcd]*': No such file or directory
sysadmin@localhost:~$ ls -d /etc/[abcd]*
/etc/adduser.conf            /etc/binfmt.d             /etc/crontab
/etc/alternatives            /etc/ca-certificates      /etc/dbus-1
/etc/apm                     /etc/ca-certificates.conf /etc/debconf.conf
/etc/apparmor                /etc/calendar             /etc/debian_version
/etc/apparmor.d              /etc/console-setup        /etc/default
/etc/apt                     /etc/cron.d               /etc/deluser.conf
/etc/bash.bashrc             /etc/cron.daily           /etc/depmod.d
/etc/bash_completion         /etc/cron.hourly          /etc/dhcp
/etc/bind                    /etc/cron.monthly         /etc/dpkg
/etc/bindresvport.blacklist  /etc/cron.weekly
sysadmin@localhost:~$
```



Chapter 8 ：



LAB-8 :

```
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 6.12.90+deb13-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.


This lab has two user accounts (username :: password )

   root     :: netlab123
   sysadmin :: netlab123

Press the [Enter] key to begin...

sysadmin@localhost:~$ echo *
Desktop Documents Downloads Music Pictures Public Templates Videos
sysadmin@localhost:~$ echo D*
```

```
sysadmin@localhost:~$ echo D*
Desktop Documents Downloads
sysadmin@localhost:~$ echo P*
Pictures Public
sysadmin@localhost:~$ echo *s
Documents Downloads Pictures Templates Videos
sysadmin@localhost:~$ echo D*n*s
Documents Downloads
sysadmin@localhost:~$ echo ??????
Public Videos
sysadmin@localhost:~$ echo D????????
Documents Downloads
sysadmin@localhost:~$ echo ?????*s
Documents Downloads Pictures Templates Videos
sysadmin@localhost:~$ echo [DP]*
Desktop Documents Downloads Pictures Public
sysadmin@localhost:~$ echo [!DP]*
Music Templates Videos
sysadmin@localhost:~$ echo [D-P]*
Desktop Documents Downloads Music Pictures Public
sysadmin@localhost:~$ echo [!D-P]*
Templates Videos
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

```
sysadmin@localhost:~$ ls
Desktop    Downloads  Pictures  Templates  hosts
Documents  Music      Public    Videos
sysadmin@localhost:~$ rm hosts
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
sysadmin@localhost:~$ cp -v /etc/hosts hosts
'/etc/hosts' -> 'hosts'
sysadmin@localhost:~$ ls
Desktop    Downloads  Pictures  Templates  hosts
Documents  Music      Public    Videos
sysadmin@localhost:~$ rm hosts
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
sysadmin@localhost:~$ cp -v /etc/hosts
cp: missing destination file operand after '/etc/hosts'
Try 'cp --help' for more information.
sysadmin@localhost:~$ cp -v /etc/hosts .
'/etc/hosts' -> './hosts'
sysadmin@localhost:~$ ls
Desktop    Downloads  Pictures  Templates  hosts
Documents  Music      Public    Videos
sysadmin@localhost:~$ rm hosts
sysadmin@localhost:~$ ls
```

```
Desktop    Downloads  Pictures  Templates  hosts
Documents  Music      Public    Videos
sysadmin@localhost:~$ rm hosts
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
sysadmin@localhost:~$ cd /etc
sysadmin@localhost:/etc$ ls -l hosts
-rw-r--r-- 1 root root 172 Oct 12 15:14 hosts
sysadmin@localhost:/etc$ cp -p hosts /home/sysadmin
sysadmin@localhost:/etc$ cd
sysadmin@localhost:~$ ls -l hosts
-rw-r--r-- 1 sysadmin sysadmin 172 Oct 12 15:14 hosts
sysadmin@localhost:~$ rm hosts
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
sysadmin@localhost:~$ cp -p /etc/hosts ~
sysadmin@localhost:~$ cp hosts newname
sysadmin@localhost:~$ ls -l hosts newnme
ls: cannot access 'newnme': No such file or directory
-rw-r--r-- 1 sysadmin sysadmin 172 Oct 12 15:14 hosts
sysadmin@localhost:~$ cp hosts newname
sysadmin@localhost:~$ ls -l hosts newname
-rw-r--r-- 1 sysadmin sysadmin 172 Oct 12 15:14 hosts
-rw-r--r-- 1 sysadmin sysadmin 172 Oct 12 15:24 newname
```

```
sysadmin@localhost:~$ cp hosts newname
sysadmin@localhost:~$ ls -l hosts newname
-rw-r--r-- 1 sysadmin sysadmin 172 Oct 12 15:14 hosts
-rw-r--r-- 1 sysadmin sysadmin 172 Oct 12 15:24 newname
sysadmin@localhost:~$ rm hosts newname
sysadmin@localhost:~$
sysadmin@localhost:~$ mkdir Myetc
sysadmin@localhost:~$ cp -R /etc/udev Myetc
sysadmin@localhost:~$ ls -l Myetc
total 0
drwxr-xr-x 4 sysadmin sysadmin 68 Oct 12 15:25 udev
sysadmin@localhost:~$ ls -lR Myetc
Myetc:
total 0
drwxr-xr-x 4 sysadmin sysadmin 68 Oct 12 15:25 udev

Myetc/udev:
total 4
drwxr-xr-x 2 sysadmin sysadmin  10 Oct 12 15:25 hwdb.d
drwxr-xr-x 2 sysadmin sysadmin  62 Oct 12 15:25 rules.d
-rw-r--r-- 1 sysadmin sysadmin 218 Oct 12 15:25 udev.conf

Myetc/udev/hwdb.d:
total 0
```

```
Myetc:
total 0
drwxr-xr-x 4 sysadmin sysadmin 68 Oct 12 15:25 udev

Myetc/udev:
total 4
drwxr-xr-x 2 sysadmin sysadmin  10 Oct 12 15:25 hwdb.d
drwxr-xr-x 2 sysadmin sysadmin  62 Oct 12 15:25 rules.d
-rw-r--r-- 1 sysadmin sysadmin 218 Oct 12 15:25 udev.conf

Myetc/udev/hwdb.d:
total 0

Myetc/udev/rules.d:
total 8
-rw-r--r-- 1 sysadmin sysadmin  306 Oct 12 15:25 70-persistent-cd.rules
-rw-r--r-- 1 sysadmin sysadmin 1157 Oct 12 15:25 README
sysadmin@localhost:~$ ls
Desktop    Downloads  Myetc     Public     Videos
Documents  Music      Pictures  Templates
sysadmin@localhost:~$ re -r Mytype

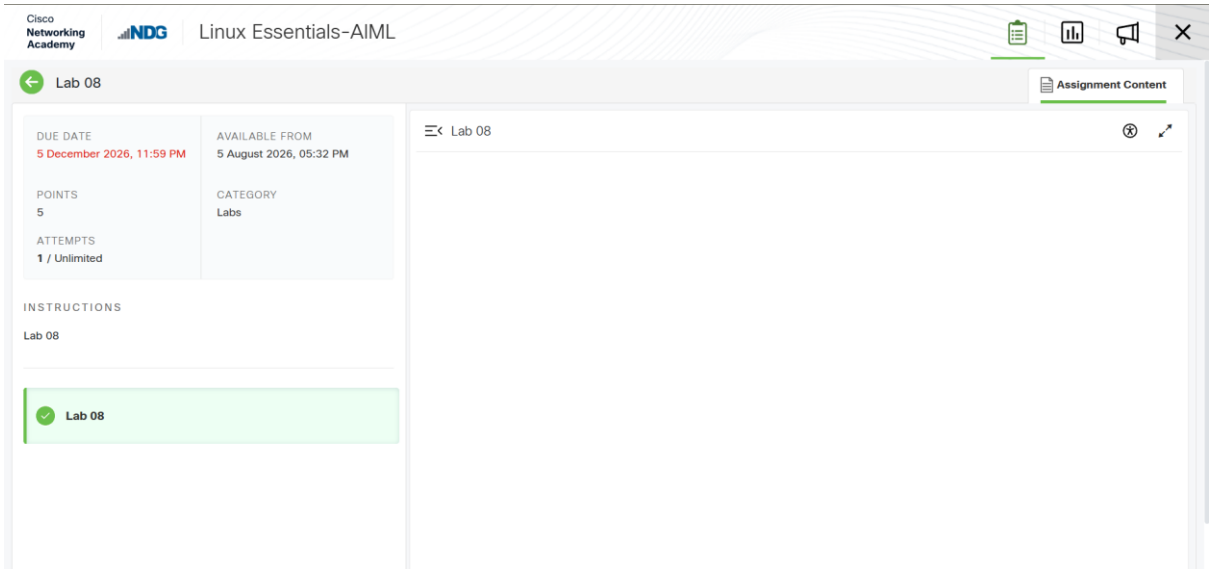
Command 're' not found, but can be installed with:
```

```
sysadmin@localhost:~$ re -r Mytype

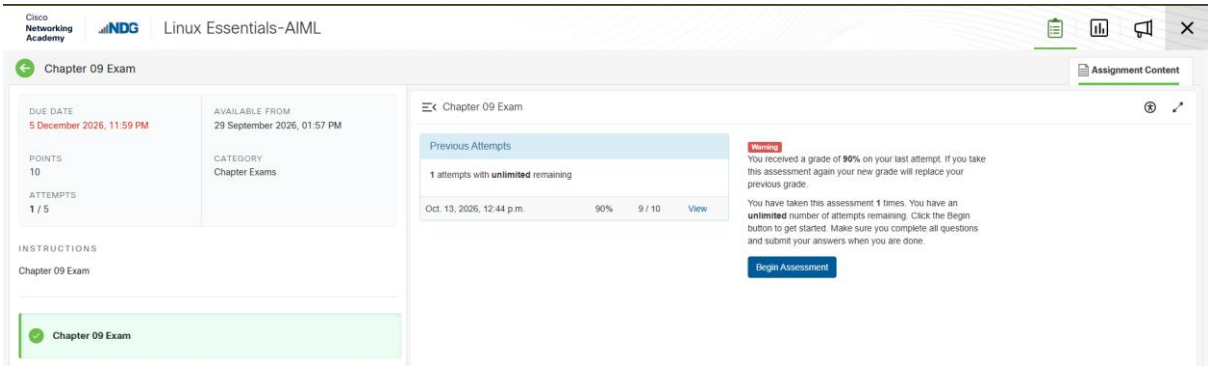
Command 're' not found, but can be installed with:

sudo apt install re

sysadmin@localhost:~$ rm -r Mytype
rm: cannot remove 'Mytype': No such file or directory
sysadmin@localhost:~$
sysadmin@localhost:~$ ls
Desktop    Downloads  Myetc     Public     Videos
Documents  Music      Pictures  Templates
sysadmin@localhost:~$ rm -r Myetc
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
sysadmin@localhost:~$ touch premove
sysadmin@localhost:~$ ls
Desktop    Downloads  Pictures  Templates  premove
Documents  Music      Public    Videos
sysadmin@localhost:~$ mv premove postmove
sysadmin@localhost:~$ ls
Desktop    Downloads  Pictures  Templates  postmove
Documents  Music      Public    Videos
sysadmin@localhost:~$
```



Chapter 9 ：



LAB-9 :

```
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 6.12.90+deb13-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.


This lab has two user accounts (username :: password )

   root     :: netlab123
   sysadmin :: netlab123

Press the [Enter] key to begin...

sysadmin@localhost:~$ cd
sysadmin@localhost:~$ mkdir
mkdir: missing operand
```

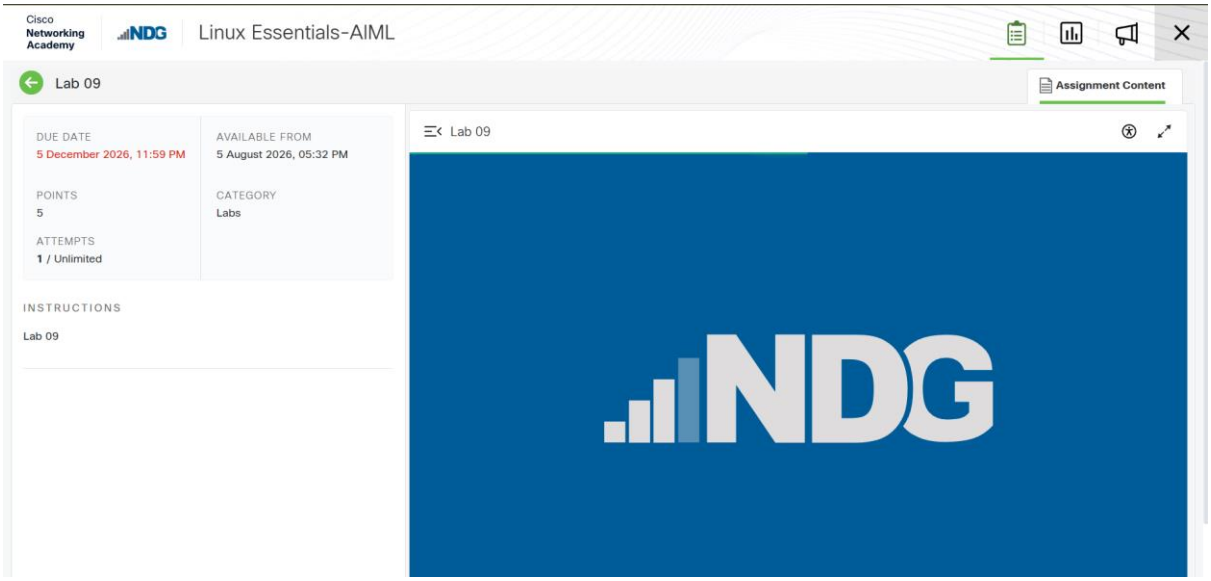
```
sysadmin@localhost:~$ cd
sysadmin@localhost:~$ mkdir mybackups
sysadmin@localhost:~$ tar -cvf mybackups/udev.tar /etc/udev
tar: Removing leading `/' from member names
/etc/udev/
/etc/udev/hwdb.d/
/etc/udev/rules.d/
/etc/udev/rules.d/70-persistent-cd.rules
/etc/udev/rules.d/README
/etc/udev/udev.conf
sysadmin@localhost:~$ ls mybackups
udev.tar
sysadmin@localhost:~$ tar -tvf mybackups/udev.tar
drwxr-xr-x root/root         0 2021-02-08 15:07 etc/udev/
drwxr-xr-x root/root         0 2021-01-06 21:04 etc/udev/hwdb.d/
drwxr-xr-x root/root         0 2021-02-08 15:08 etc/udev/rules.d/
-rw-r--r-- root/root       306 2021-02-08 15:08 etc/udev/rules.d/70-persistent-c
d.rules
-rw-r--r-- root/root      1157 2021-02-08 15:08 etc/udev/rules.d/README
-rw-r--r-- root/root       218 2021-02-08 15:08 etc/udev/udev.conf
sysadmin@localhost:~$ tar -zcvf mybackups/udev.tar.gz /etc/udev
tar: Removing leading `/' from member names
/etc/udev/
/etc/udev/hwdb.d/
```

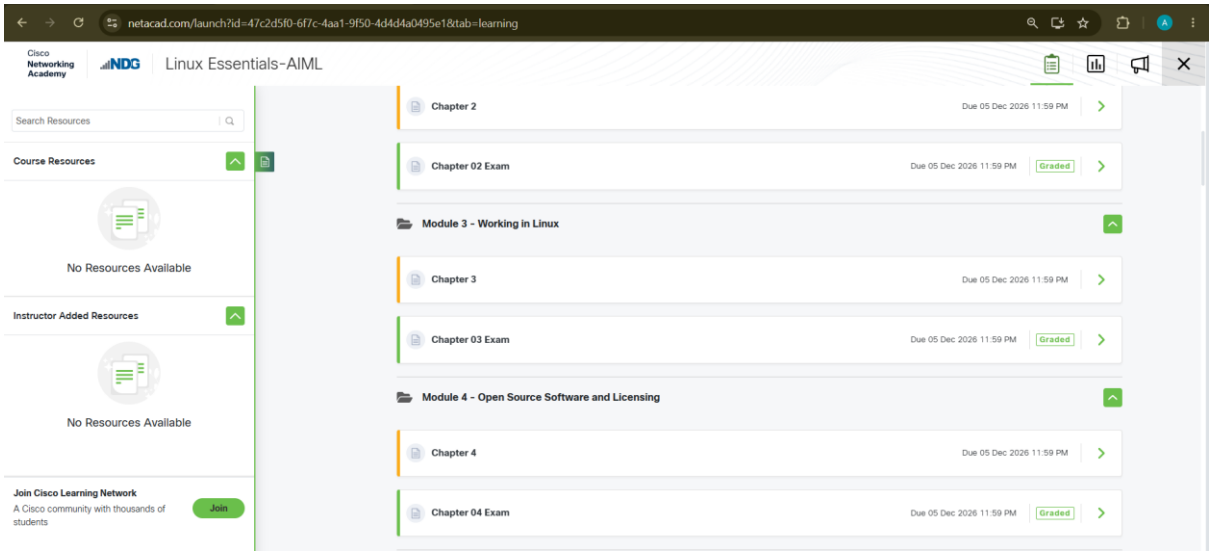
```
/etc/udev/rules.d/70-persistent-cd.rules
/etc/udev/rules.d/README
/etc/udev/udev.conf
sysadmin@localhost:~$ ls -lh mybackups/
total 16K
-rw-rw-r-- 1 sysadmin sysadmin  10K Oct 13 04:53 udev.tar
-rw-rw-r-- 1 sysadmin sysadmin 1.2K Oct 13 04:54 udev.tar.gz
sysadmin@localhost:~$ cd mybackups
sysadmin@localhost:~/mybackups$ ls
udev.tar  udev.tar.gz
sysadmin@localhost:~/mybackups$ tar -xvf udev.tar.gz
etc/udev/
etc/udev/hwdb.d/
etc/udev/rules.d/
etc/udev/rules.d/70-persistent-cd.rules
etc/udev/rules.d/README
etc/udev/udev.conf
sysadmin@localhost:~/mybackups$ ls
etc  udev.tar  udev.tar.gz
sysadmin@localhost:~/mybackups$ ls etc
udev
sysadmin@localhost:~/mybackups$ ls etc/udev
hwdb.d  rules.d  udev.conf
sysadmin@localhost:~/mybackups$ ls etc/udev/rules.d
```

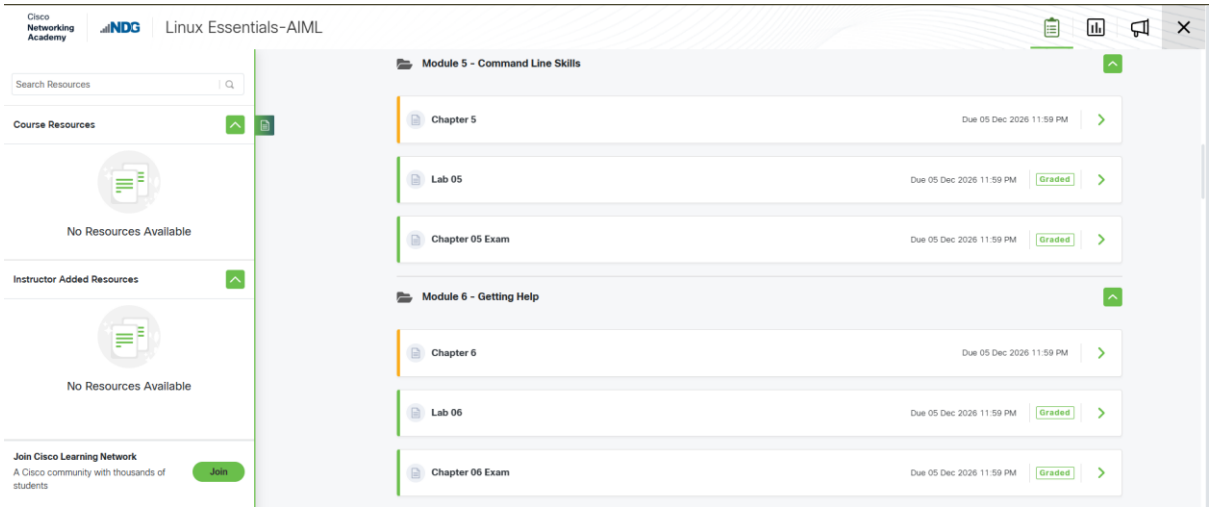
```
udev
sysadmin@localhost:~/mybackups$ ls etc/udev
hwdb.d  rules.d  udev.conf
sysadmin@localhost:~/mybackups$ ls etc/udev/rules.d
70-persistent-cd.rules  README
sysadmin@localhost:~/mybackups$ tar -rvf udev.tar /etc/hosts
tar: Removing leading `/' from member names
/etc/hosts
sysadmin@localhost:~/mybackups$ tar -tav udev.tar
tar: Refusing to read archive contents from terminal (missing -f option?)
tar: Error is not recoverable: exiting now
sysadmin@localhost:~/mybackups$ cp /usr/share/dict/words
cp: missing destination file operand after '/usr/share/dict/words'
Try 'cp --help' for more information.
sysadmin@localhost:~/mybackups$ cp /usr/share/dict/words .
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct 13 04:57 words
sysadmin@localhost:~/mybackups$ gzip words
sysadmin@localhost:~/mybackups$ ls -l words
ls: cannot access 'words': No such file or directory
sysadmin@localhost:~/mybackups$ ls -l words.gz
-rw-r--r-- 1 sysadmin sysadmin 259983 Oct 13 04:57 words.gz
sysadmin@localhost:~/mybackups$ ls -l words.gz
-rw-r--r-- 1 sysadmin sysadmin 259983 Oct 13 04:57 words.gz
```

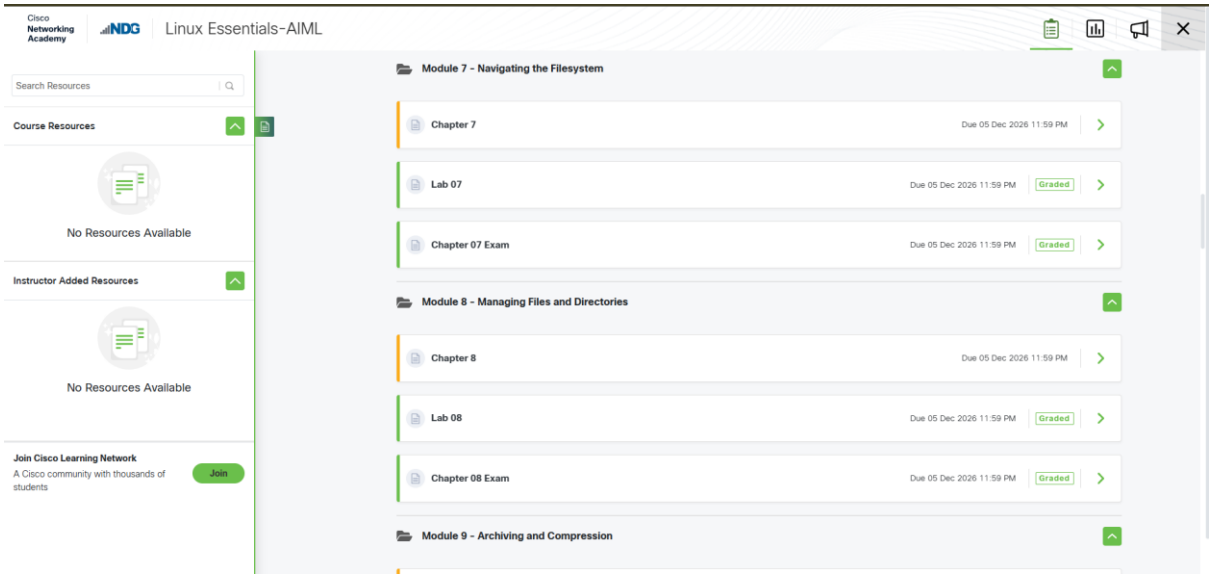
```
-rw-r--r-- 1 sysadmin sysadmin 259983 Oct 13 04:57 words.gz
sysadmin@localhost:~/mybackups$ gunzip words.gz
sysadmin@localhost:~/mybackups$ ls -l worda
ls: cannot access 'worda': No such file or directory
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct 13 04:57 words
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct 13 04:57 words
sysadmin@localhost:~/mybackups$ bzip2 words
sysadmin@localhost:~/mybackups$ ls -l words.bz2
-rw-r--r-- 1 sysadmin sysadmin 345560 Oct 13 04:57 words.bz2
sysadmin@localhost:~/mybackups$ ls -l words
ls: cannot access 'words': No such file or directory
sysadmin@localhost:~/mybackups$ ls -l words
ls: cannot access 'words': No such file or directory
sysadmin@localhost:~/mybackups$ xz words
xz: words: No such file or directory
sysadmin@localhost:~/mybackups$ bunzip2 words.bz2
sysadmin@localhost:~/mybackups$ ls -l words
-rw-r--r-- 1 sysadmin sysadmin 971578 Oct 13 04:57 words
sysadmin@localhost:~/mybackups$ ls -l words
```

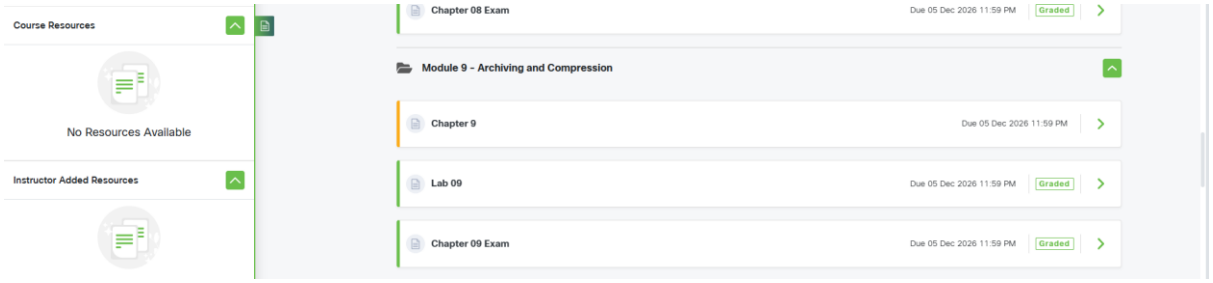
```
-rw-rw-r-- 1 sysadmin sysadmin 260119 Oct 13 05:06 words.zip
sysadmin@localhost:~/mybackups$ zip -r udev.zip /etc/udev
  adding: etc/udev/ (stored 0%)
  adding: etc/udev/hwdb.d/ (stored 0%)
  adding: etc/udev/rules.d/ (stored 0%)
  adding: etc/udev/rules.d/70-persistent-cd.rules (deflated 29%)
  adding: etc/udev/rules.d/README (deflated 50%)
  adding: etc/udev/udev.conf (deflated 24%)
sysadmin@localhost:~/mybackups$ ls -l udev.zip
-rw-rw-r-- 1 sysadmin sysadmin 2000 Oct 13 05:07 udev.zip
sysadmin@localhost:~/mybackups$ unzip -l udev.zip
Archive:  udev.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  2021-02-08 15:07   etc/udev/
        0  2021-01-06 21:04   etc/udev/hwdb.d/
        0  2021-02-08 15:08   etc/udev/rules.d/
      306  2021-02-08 15:08   etc/udev/rules.d/70-persistent-cd.rules
     1157  2021-02-08 15:08   etc/udev/rules.d/README
      218  2021-02-08 15:08   etc/udev/udev.conf
---------                     -------
     1681                     6 files
sysadmin@localhost:~/mybackups$ rm -r etc
sysadmin@localhost:~/mybackups$
```




netacad.com/launch?id=47c2d5f0-6f7c-4aa1-9f50-4d4d4a0495e1&tab=learning
Cisco Networking Academy
NDG
Linux Essentials-AIML
Search Resources
Course Resources
No Resources Available
Instructor Added Resources
No Resources Available
Join Cisco Learning Network
A Cisco community with thousands of students
Join
Chapter 2
Due 05 Dec 2026 11:59 PM
Chapter 02 Exam
Due 05 Dec 2026 11:59 PM
Graded
Module 3 - Working in Linux
Chapter 3
Due 05 Dec 2026 11:59 PM
Chapter 03 Exam
Due 05 Dec 2026 11:59 PM
Graded
Module 4 - Open Source Software and Licensing
Chapter 4
Due 05 Dec 2026 11:59 PM
Chapter 04 Exam
Due 05 Dec 2026 11:59 PM
Graded


Cisco Networking Academy
NDG
Linux Essentials-AIML
Search Resources
Course Resources
No Resources Available
Instructor Added Resources
No Resources Available
Join Cisco Learning Network
A Cisco community with thousands of students
Join
Module 5 - Command Line Skills
Chapter 5
Due 05 Dec 2026 11:59 PM
Lab 05
Due 05 Dec 2026 11:59 PM
Graded
Chapter 05 Exam
Due 05 Dec 2026 11:59 PM
Graded
Module 6 - Getting Help
Chapter 6
Due 05 Dec 2026 11:59 PM
Lab 06
Due 05 Dec 2026 11:59 PM
Graded
Chapter 06 Exam
Due 05 Dec 2026 11:59 PM
Graded


Cisco Networking Academy
NDG
Linux Essentials-AIML
Search Resources
Course Resources
No Resources Available
Instructor Added Resources
No Resources Available
Join Cisco Learning Network
A Cisco community with thousands of students
Join
Module 7 - Navigating the Filesystem
Chapter 7
Due 05 Dec 2026 11:59 PM
Lab 07
Due 05 Dec 2026 11:59 PM
Graded
Chapter 07 Exam
Due 05 Dec 2026 11:59 PM
Graded
Module 8 - Managing Files and Directories
Chapter 8
Due 05 Dec 2026 11:59 PM
Lab 08
Due 05 Dec 2026 11:59 PM
Graded
Chapter 08 Exam
Due 05 Dec 2026 11:59 PM
Graded
Module 9 - Archiving and Compression



Midterm Exam :

